

CHI-SHENG CHEN

EMBEDDED SYSTEM DEVELOPER

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GitHub: [rota1001](https://github.com/rota1001)

Education

National Cheng Kung University (Sep 2023 – Jun 2027)

- B.S. in Computer Science and Engineering
- Academic Excellence Award (1st x 3, Top 5% x 4, GPA 4.28/4.3)

Skills

- Languages: C, C++, python
- Linux: Linux Kernel, System Integration (bootloader, libc, rootfs, busybox,...), Driver
- Embedded System: Bare Metal, GDB, QEMU, Protocols (UART, SPI, I2C, PCIe, VGA, ...), Arm Cortex-M
- Server Software: EDK2, DXE/PEI Driver, IPMI, BMC

Experience

Google Software Engineering Intern - x86 Platform SW Team (Jun 2026 - Sep 2026)

- Reduced boot time by **16.5%** by improving the **PCIe** memory resource allocation strategy in **EDK2**
- Implemented **DXE and PEI drivers** in EDK2 and IPMI OEM command on **BMC**

Presentations

- Embedded Linux Conference North America 2026 — *Optimize Linux Kernel to Fit Microcontroller with 1MB RAM*^[1]
- HITCON 2025 — *A Kernel Rootkit That Works Without Kallsyms*^[2]
- SITCON 2026 — *STM32 Bare-metal DOOM Challenge*^[3]

Side Projects

Run Linux 7.0 on STM32H750 (Cortex-M7, no-MMU, 1MiB SRAM)

Description: Run **Linux 7.0** on a microcontroller with only **1MiB SRAM**

Links: Github^[4] Hackmd^[5]

- Developed a **QEMU SoC model**, enabling kernel debugging and memory profiling
- Integrated a minimal Linux system consisting of a **bootloader**, **romfs**, **uClibc-ng**, and **BusyBox**
- Booted the Linux kernel to BusyBox while retaining **256 KiB of free RAM**
- Configured user programs in **FDPIC** format for **Execute-in-Place (XIP)**, consuming zero RAM for the text segment

A Feature-Rich Linux integration on STM32F429 (Cortex-M4, no-MMU, 8MiB SDRAM)

Description: Integrate a Linux system capable of running a **cellphone prototype** with **8MiB SDRAM** and **2MiB Flash**

Links: Github^[6] Hackmd^[7]

- Integrated a minimal Linux system consisting of a **bootloader**, **romfs**, **rootfs (ext2)**, **uClibc-ng**, and **BusyBox**
- Tuned kernel configurations to enable **LCD**, **MMC**, **I2C**, and **SPI** drivers
- Leveraged **tslib** with the Linux **input subsystem** to enable touchscreen support
- Deployed the **LVGL** graphics library on **DRM**
- Ported an interactive cellphone UI prototype featuring touch gestures and a swipe-to-type virtual keyboard

Port DOOM to STM32H750 (SITCON 2026)

Description: Run 3D game on STM32H750 with **only 1MB SRAM** and **without STM32 HAL**.

Links: Github[8] Hackmd[9]

- Pure register-level programming using **MMIO**
- Used **LTDC** and DAC to output 640x480@64Hz **VGA** signal with only 320x200 bytes framebuffer
- XIP code execution on **QSPI** external flash
- Modified the memory management system to support non-contiguous memory allocation in DOOM
- Implemented libc support for DOOM

Linux Kernel Instrumentation Framework (HITCON 2025)

Description: Kernel module for intercepting kernel execution paths **without kallsyms or kprobes**. Presented at *HITCON 2025*.

Links: Github[10] Hackmd[11]

- Inspected and modified **procfs** file operations by parsing its internal tree structure
- Resolved syscall function addresses by analyzing the **syscall handler** without relying on exported syscall table
- Performed runtime kernel patching
- Achieved process concealment by manipulating the internal **PID hash** structure used by the scheduler
- Perform **ELF injection** to load the kernel module during systemd startup

Microcomputer with Custom OS and Peripheral Integration

Description: Built a microcomputer on the bread board using PIC18F integrating keyboard, SRAM, UART, and USB storage.

Links: Github[12] Hackmd[13]

- Developed **USB Host driver** for CH375 supporting **HID keyboard** and USB flash drive
- Built external memory interface enabling **1-byte-per-instruction** block access using **only an 8-bit GPIO** with latch/counter/D-trigger
- Implemented **DMA mechanism** using 512KB SRAM as disk cache

Competitions

Programming Contests

- 2025 ICPC Taichung Regional — Bronze Medal
- 2024 ICPC Taichung Regional — Bronze Medal

CTF Contests (Security)

- HITCON CTF 2025 Quals — Ranked 2nd (Taiwan) / Ranked 22nd (Global), write-up[14]
- Crypto CTF 2025 — Ranked 38th (Solo, Global), write-up[15]

Links

- [1] <https://osselcna2026.sched.com/event/2JQvE>
- [2] <https://hitcon.org/2025/en-US/agenda/f679c826-4c37-4989-bf27-6e44ea4726ce/>
- [3] <https://sitcon.org/2026/agenda/91836d>
- [4] <https://github.com/rota1001/stm32h7-linux>
- [5] <https://hackmd.io/@rota1001/stm32h750-linux>
- [6] <https://github.com/rota1001/stm32f429-linux>
- [7] <https://hackmd.io/@rota1001/linux-stm32f429>
- [8] <https://github.com/rota1001/stm32h7-baremetal-doom>
- [9] <https://hackmd.io/@rota1001/stm32-doom>

- [10] <https://github.com/rota1001/ksymless>
- [11] https://hackmd.io/@sysprog/r1l3i4_Zge
- [12] <https://github.com/beautiful-fruit/picos>
- [13] <https://hackmd.io/@weiso131/picos>
- [14] <https://hackmd.io/@rota1001/hitcon-ctf-quals-2025>
- [15] <https://hackmd.io/@rota1001/crypto-ctf-2025>